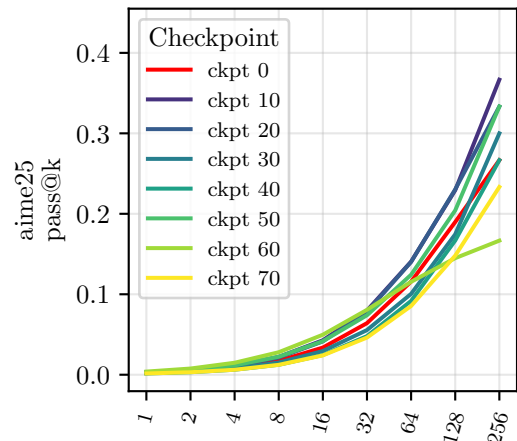
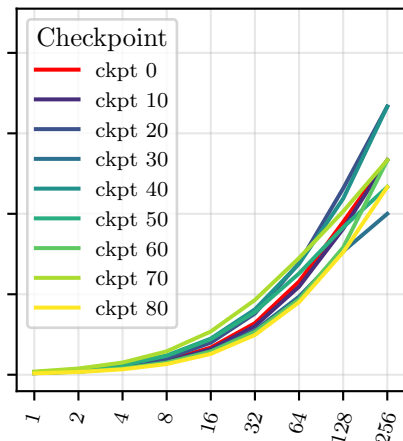


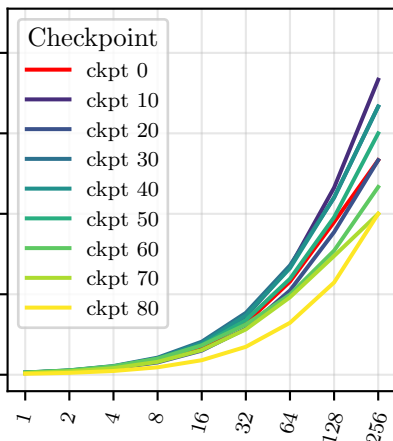
EuroLLM-9B-Instruct, GRPO



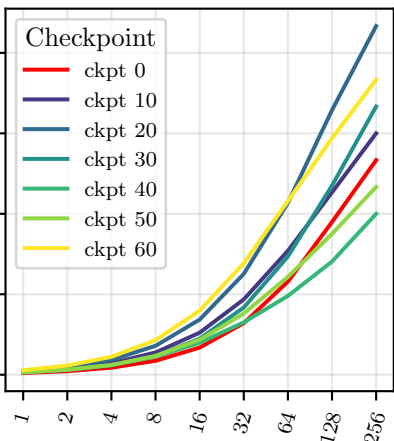
EuroLLM-9B-Instruct, Pass@k Training



EuroLLM-9B-Instruct, GRPO-S

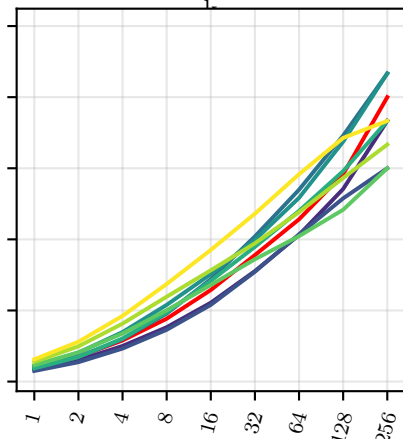


EuroLLM-9B-Instruct, KL-Cov



A line graph showing the relationship between the number of passes (k) and the pass@k score for the bruno25 dataset. The x-axis represents k (1, 2, 4, 8, 16, 32, 64, 128, 256) on a logarithmic scale. The y-axis represents the pass@k score (0.0 to 0.5). Five models are compared: GPT-4o (red), Gemini-2.0 Flash (green), Gemini-2.0 Pro (blue), Gemini-2.0 Pro Vision (purple), and Gemini-2.0 Pro Vision Thinking (orange). All models show an increasing trend in pass@k as k increases. Gemini-2.0 Pro Vision Thinking generally achieves the highest pass@k scores, followed by Gemini-2.0 Pro Vision, Gemini-2.0 Pro, Gemini-2.0 Flash, and GPT-4o.

k	GPT-4o	Gemini-2.0 Flash	Gemini-2.0 Pro	Gemini-2.0 Pro Vision	Gemini-2.0 Pro Vision Thinking
1	0.02	0.02	0.02	0.02	0.02
2	0.03	0.03	0.03	0.03	0.03
4	0.05	0.05	0.05	0.05	0.05
8	0.08	0.08	0.08	0.08	0.08
16	0.12	0.12	0.12	0.12	0.12
32	0.18	0.18	0.18	0.18	0.18
64	0.25	0.25	0.25	0.25	0.25
128	0.30	0.30	0.30	0.30	0.30
256	0.35	0.35	0.35	0.35	0.35

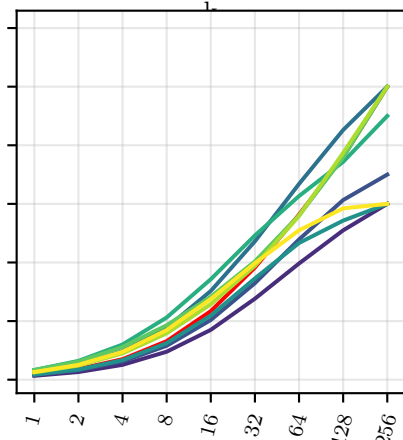


n	alpha = 0.1 (green)	alpha = 0.2 (blue)	alpha = 0.3 (red)	alpha = 0.4 (cyan)	alpha = 0.5 (yellow)
1	1.0	1.0	1.0	1.0	1.0
2	1.5	1.4	1.3	1.2	1.1
4	2.5	2.2	2.0	1.8	1.6
8	4.0	3.5	3.2	2.8	2.5
16	6.0	5.0	4.5	4.0	3.5
32	9.0	7.5	6.5	5.5	4.5
64	13.0	11.0	9.5	8.0	6.5
128	18.0	15.0	13.0	11.0	9.0
256	25.0	21.0	18.0	15.0	12.0

n	L1 (red)	L2 (blue)	L3 (yellow)	L4 (green)
1	10	12	14	16
2	15	18	20	22
4	22	26	28	30
8	30	35	38	40
16	40	45	48	50
32	50	55	58	60
64	65	70	72	75
128	80	85	88	90
256	95	100	105	110

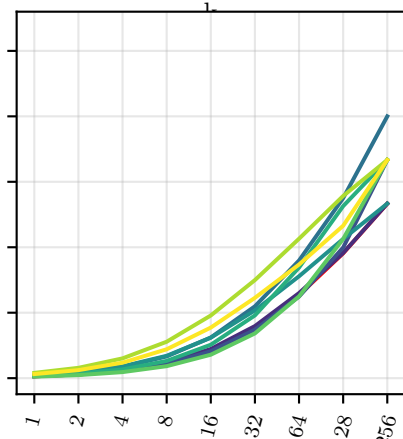
A line graph showing the performance of various models on the cmimc25 dataset, measured by the pass@k metric. The x-axis represents k (1, 2, 4, 8, 16, 32, 64, 128, 256) and the y-axis represents cmimc25 pass@k (0.00 to 0.30). The models are represented by different colored lines: red (top), blue, green, yellow, and light green (bottom). All models show an increasing trend in pass@k as k increases, with the red model consistently performing the best.

k	Red	Blue	Green	Yellow	Light Green
1	0.005	0.005	0.005	0.005	0.005
2	0.010	0.010	0.010	0.010	0.010
4	0.025	0.025	0.025	0.025	0.025
8	0.045	0.045	0.045	0.045	0.045
16	0.075	0.075	0.075	0.075	0.075
32	0.110	0.110	0.110	0.110	0.110
64	0.145	0.145	0.145	0.145	0.145
128	0.195	0.195	0.195	0.195	0.195
256	0.250	0.230	0.200	0.180	0.150



N	Green	Blue	Red	Yellow	Dark Blue
1	0.05	0.05	0.05	0.05	0.05
2	0.08	0.08	0.08	0.08	0.08
4	0.15	0.15	0.15	0.15	0.15
8	0.25	0.25	0.25	0.25	0.25
16	0.40	0.40	0.40	0.40	0.40
32	0.60	0.60	0.60	0.60	0.60
64	0.80	0.80	0.80	0.80	0.80
128	1.00	1.00	1.00	0.95	0.95
256	1.20	1.20	1.20	0.95	0.95

Figure 1 is a line graph showing the pass@k performance of the hmm-t25 model for different numbers of samples (k). The x-axis represents k (1, 2, 4, 8, 16, 32, 64, 128, 256) and the y-axis represents pass@k (0.00 to 0.25). Five curves are shown, representing different model sizes or configurations, all showing an increasing trend in pass@k as k increases.



n	L1 (red)	L2 (blue)	L3 (yellow)	L4 (green)
1	0.5	0.5	0.5	0.5
2	0.6	0.6	0.6	0.6
4	0.8	0.8	0.8	0.8
8	1.2	1.3	1.4	1.5
16	2.0	2.2	2.4	2.6
32	3.2	3.6	4.0	4.4
64	5.0	5.6	6.0	6.4
128	7.5	8.4	8.8	9.2
256	11.0	12.8	13.2	14.0

k	GPT-4o (red)	Gemini-1.5 Pro (green)	Gemini-1.5 Flash (blue)	Gemini-1.5 Pro Experimental (purple)	Gemini-1.5 Flash Experimental (yellow)
1	0.28	0.26	0.25	0.24	0.23
2	0.38	0.36	0.35	0.34	0.33
4	0.48	0.46	0.45	0.44	0.43
8	0.58	0.56	0.55	0.54	0.53
16	0.68	0.66	0.65	0.64	0.63
32	0.75	0.73	0.72	0.71	0.70
64	0.80	0.78	0.77	0.76	0.75
128	0.83	0.81	0.80	0.79	0.78
256	0.86	0.84	0.83	0.82	0.81

